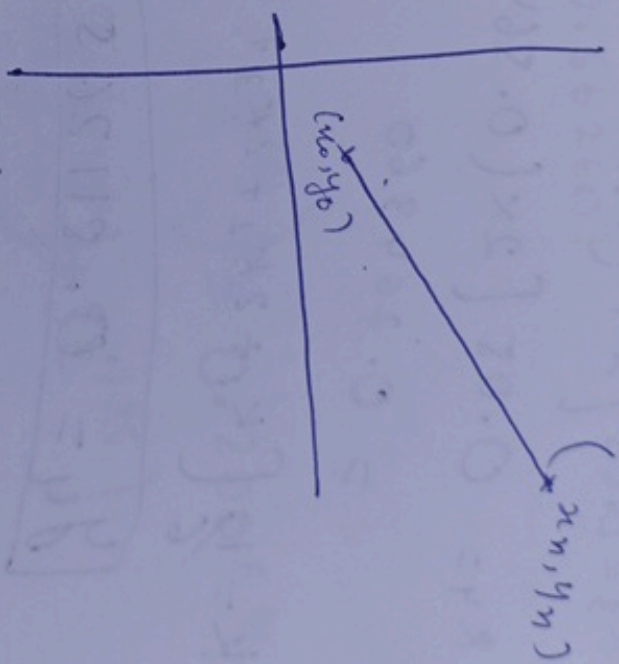


Unit-4

errors cannot be negative.

Curve Fitting

x_0	y_0	e_1
x_1	y_1	e_2
x_2	y_2	e_3
$\vdots$	$\vdots$	$\vdots$
x_n	y_n	e_n



Using Principal of least square

$$(e_1)^2 + (e_2)^2 + \dots + (e_n)^2 \rightarrow \sum_{i=1}^n e_i^2 / n = \text{Error for Curve } (e)$$

$$\sum y = 16.9$$

$$(50)$$

$$\frac{1}{2} \frac{3}{4} \frac{10}{10}$$

① Principle of least square:

① Fit a line $[Y = ax + b]$ (degree 1)

Normal Equations

$$\begin{aligned} \sum y &= na + b \sum x \\ \sum xy &= a \sum x + b \sum x^2 \end{aligned}$$

$n = \text{no. of observations.}$

after $(y = a + bx)$

x x_0 x_1 ...	y y_0 y_1 ...	xy xy_0 xy_1 ...	x^2 x_0^2 x_1^2 ...	x^2	$y'(\text{approx})$	error
$\frac{\sum x}{n}$	$\frac{\sum y}{n}$	$\frac{\sum xy}{n}$	$\frac{\sum x^2}{n}$			
Ques	x	y	xy	x^2	$y'(\text{approx})$	error
	0	1.0	0	0	0.72	10.28
	1	1.8	1.8	1	2.05	10.25
	2	3.3	6.6	4	3.38	1.08
	3	4.5	13.5	9	4.71	10.21
	4	6.3	25.2	16	6.04	10.26
	10	16.9	47.1	30		1.08

$$\sum y = na + b \sum x$$

$$16.9 = 5a + 10b$$

$$\sum xy = a \sum x + b \sum x^2$$

$$47.1 = 10a + 30b$$

$$\begin{aligned} (5a + 10b = 16.9) \times 2 &= 10a + 20b = 33.8 \\ 10a + 30b &= 47.1 \\ \hline &10b = 13.3 \\ &b = 1.33 \end{aligned}$$

$$\begin{aligned} b &= 1.33 \\ a &= 0.72 \end{aligned}$$

$$y = a + bx$$

$$y = .72 + 1.33x$$

$$\text{Average error} = \frac{\sum e}{n} = \frac{1.08}{5} = 0.216$$

Ques (ii) Fit a Parabola (curve of 2nd degree)

$$y = a + bx + cx^2$$

Normal Equations are:

$$\sum y = na + b\sum x + c\sum x^2$$

$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3$$

$$\sum x^2y = a\sum x^2 + b\sum x^3 + c\sum x^4$$

x	y	xy	x ²	x ³	x ⁴	x ² y
0	1.0	0	0	0	0	0
1	1.8	1.8	1	1	1	1.8
2	3.3	6.6	4	8	16	13.2
3	4.5	13.5	9	27	81	40.5
4	6.3	25.2	16	64	256	100.8
<u>10</u>	<u>16.9</u>	<u>47.1</u>	<u>30</u>	<u>100</u>	<u>354</u>	<u>156.3</u>

$$16.9 = 5a + 10b + 30c$$

$$47.1 = 10a + 30b + 100c$$

$$156.3 = 30a + 100b + 354c$$

Using Gauss Elimination

$$\begin{bmatrix} 5 & 10 & 30 & 16.9 \\ 10 & 30 & 100 & 47.1 \\ 30 & 100 & 35 & 156.3 \end{bmatrix}$$

$$\frac{a_2}{a_1} = 2$$

$$\frac{a_3}{a_1} = 6$$

$$R_2 \rightarrow R_2 - \frac{a_2}{a_1} R_1, R_3 \rightarrow R_3 - \frac{a_3}{a_1} R_1$$

$$\begin{bmatrix} 5 & 10 & 30 & 16.9 \\ 0 & 10 & 40 & 13.3 \\ 0 & 40 & 174 & 54.9 \end{bmatrix}$$

$$\frac{b_3}{b_2} = 4$$

$$R_3 \rightarrow R_3 - \frac{b_3}{b_2} R_2$$

$$\begin{bmatrix} 5 & 10 & 30 & 16.9 \\ 0 & 10 & 40 & 13.3 \\ 0 & 0 & 14 & 1.7 \end{bmatrix}$$

Backward Substitution

$$\begin{bmatrix} a = 0.962856 \\ b = 0.844288 \\ c = \frac{1.7}{14} = 0.121428 \end{bmatrix}$$

$$y = 0.962856 + 0.844288x + 0.121428x^2$$

y' (approx)

$$x_0 \quad 0.962856$$

$$x_1 \quad 1.928564$$

$$x_2 \quad 3.10855$$

$$x_3 \quad 4.52424$$

$$x_4 \quad 6.168488$$

error

$$10.0371441$$

$$10.1285641$$

$$10.1914481$$

$$10.024241$$

$$10.1315121$$

$$0.102582$$

$$\frac{\sum e}{n} = 0.102582$$

③ Polynomial Curve $\rightarrow$

$$y = ax^b$$

taking logs on both sides.

$$\log y = \log(ax^b)$$

$$\log y = \log a + \log x^b$$

$$\log y = \log a + b \log x$$

$$y = A + Bx$$

$$\sum Y = nA + B \sum X$$

$$\sum XY = A \sum X + B \sum X^2$$

Previous Equation

x	y	$X = \log_{10} x$	$Y = \log_{10} y$
0	1.0		
1	1.8		
2	3.3	0.301029	
3	4.5	0.477121	
4	6.3	0.799341	

$\log 0$ is not defined
So this data
is not applicable
for this
curve

Anti log = 10^x

like
 $x = \log_b y$

$$y = b^x$$

⑤

Ans

Exponential Curve: $y = ae^{bx}$

taking log on both sides at base e.

$$\log_e y = \log_e a + bx \log_e e$$

$\left[\log_e e = 1 \right]$ if base & log then its value = 1.
 $\left[\log_a a = 1 \right]$

$$\log_e y = \log_e a + bx$$

$\downarrow$ $\downarrow$ $\downarrow \downarrow$
 y A $+ Bx$

$$y = A + Bx$$

$$\begin{matrix} A = \log_e a \\ B = b \end{matrix} \rightarrow a = \text{Antilog}(A)$$

Ques An experiment gave the following values

$v \text{ (ft/min)}$:	350	400	500	600
$t \text{ (min)}$:	61	26	7	2.6

it is known that v and t are connected by the relation $v = at^b$, find the best possible values of a and b .

$$v = at^b$$

taking log on both side

$$\log v = \log a + b \log t$$

$\downarrow$ $\downarrow$ $\downarrow \downarrow$ (let)
 $y = A + Bx$

$$\begin{matrix} y = \log v \\ x = \log t \end{matrix}$$

~~2.54408~~

$Y = \log v$	$X = \log t$	X^2	XY
2.544068	1.785329	3.18799	4.541998
2.60206	1.414973	2.002148	3.681845
2.698970	0.845098	0.714191	2.280894
2.778151	0.414973	0.172202	1.152858
<u>10.623249</u>	<u>4.460373</u>	<u>6.076531</u>	<u>11.657595</u>

$$\sum Y = nA + B \sum X$$

$$n = 4$$

$$\sum XY = A \sum X + B \sum X^2$$

$$\sum Y = nA + B \sum X$$

$$10.623249 = 4A + 4.460373B$$

$$11.657595 = 4.460373A + 6.076531B$$

using Gauss Elimination

$$\left[\begin{array}{cc|c} 4 & 4.460373 & 10.623249 \\ 4.460373 & 6.076531 & 11.657595 \end{array} \right]$$

$$\frac{a_2}{a_1} = 1.15093$$

$$R_2 = R_2 - \frac{a_2}{a_1} R_1$$

$$\left[\begin{array}{cc|c} 4 & 4.460373 & 10.623249 \\ 0 & 1.102800 & -0.188316 \end{array} \right]$$

$$\left[\begin{array}{l} A = 2.846228 \\ B = \frac{-0.188316}{1.102800} = -0.170762 \end{array} \right]$$

$$b = B = -0.170762$$

$$a = \text{antilog}(A)$$

$$= \text{antilog}(2.846228)$$

$$= 701.823652$$

$$\text{antilog}_{10} = \text{shift} + \log$$

$$\text{antilog}_e = \text{shift} + \ln$$

$$U = 701.823652 + (-0.170762)$$

⇒ Correlation and Regression.

define relation b/w
Dependent Variable
and Independent
Variable

define amount of
relation b/w

ex:

$$y = ax + b$$

$$y = ax + by + c$$

Correlation Coefficient $\rightarrow r$ [its maximum value
= 1] or b/w 0 & 1.

$$r = \frac{\text{Cov}(x, y)}{S.D_x * S.D_y}$$

(*) Covariance - % change b/w
two variables.

(*) Standard Deviation:
deviation of a value from
its mean value.

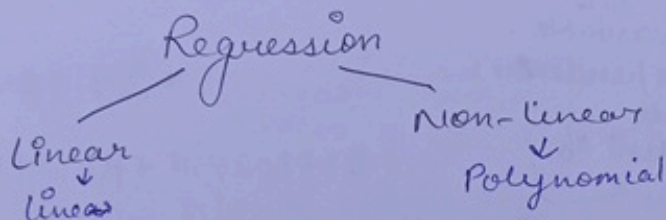
$$\text{Cov}(x, y) = \sum xy - \frac{\sum x \cdot \sum y}{n}$$

$$S.D_x = \sigma_x = \sqrt{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$S.D_y = \sigma_y = \sqrt{\sum y^2 - \frac{(\sum y)^2}{n}}$$

$$r = \frac{n \sum xy - \sum x \cdot \sum y}{\sqrt{\frac{n \sum x^2 - (\sum x)^2}{n}} \sqrt{\frac{n \sum y^2 - (\sum y)^2}{n}}}$$

$$r = \frac{n \sum xy - \sum x \cdot \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$



① Linear Regression

Lines of Regression

X on Y
or
Y on X

Regression Coefficients

b_{xy} (X on Y)

b_{yx} (Y on X)

$$b_{xy} = r \cdot \frac{\sigma_x}{\sigma_y}$$

$$b_{yx} = r \cdot \frac{\sigma_y}{\sigma_x}$$

$$b_{xy} = \frac{n \sum xy - \sum x \cdot \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}} \times \frac{\sqrt{\frac{n \sum x^2 - (\sum x)^2}{n}}}{\sqrt{\frac{n \sum y^2 - (\sum y)^2}{n}}}$$

b_{xy}

$b_{yx} =$

b_{yx}

b_{xy}

→ Line

→ Line

$$b_{xy} = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum y^2 - (\sum y)^2}$$

$$b_{yx} = \frac{n \sum xy - \sum x \cdot \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}} \times \sqrt{\frac{n \sum y^2 - (\sum y)^2}{n}}$$

$$b_{yx} = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum x^2 - (\sum x)^2}$$

$$b_{xy} * b_{yx} = r \frac{\frac{\sum y}{\sum x}}{\frac{\sum x}{\sum y}} * r \frac{\frac{\sum x}{\sum y}}{\frac{\sum y}{\sum x}}$$

$$r^2 = b_{xy} * b_{yx}$$

or

$$r = \sqrt{b_{xy} * b_{yx}}$$

→ line of y on $x \Rightarrow y = ax + b$
 $(y - \bar{y}) = b_{yx} (x - \bar{x})$

→ line of x on $y \Rightarrow x = cy + d$
 $(x - \bar{x}) = b_{xy} (y - \bar{y})$

($\bar{x}$ & $\bar{y}$ are mean values)

$$(\bar{x} = \frac{\sum x}{n}, \bar{y} = \frac{\sum y}{n})$$

Ques Find the two lines of regression & Coefficient of Correlation for the data given below.

$$n=18, \sum x=12, \sum y=18, \sum x^2=60, \sum y^2=96$$

$$\sum xy=48$$

$$b_{yx} = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum x^2 - (\sum x)^2} = \frac{18(48) - (12)(18)}{18(60) - 12 \times 12}$$

$$= .692308$$

$$b_{xy} = \frac{18(48) - (12)(18)}{18 \times 96 - 18 \times 18} = 0.461538$$

Correlation

$$[r = \sqrt{b_{yx} * b_{xy}} = \sqrt{0.692308 \times 0.461538}]$$

$$= 0.565267$$

$$\bar{x} = \frac{12}{18} = 0.667$$

$$\bar{y} = \frac{18}{18} = 1$$

→ line of y on x

$$(y - \bar{y}) = b_{yx}(x - \bar{x})$$

$$(y - 1) = 0.692308x - 0.461769$$

$$y = 0.692308x + 0.538231$$

— two line of regression.

→ line of
(x - 0.667)

$$\bar{x} =$$

Ques for two

x

6

2

10

4

8

30

b_{xy}

b_{xy}

$\sum$

$\sum$

$\sum$

$\sum$

$\sum$

\$
eleon.

→ line of x on y -

$$(x - 0.667) = 0.461538y - 0.461538$$

$$\boxed{x = 0.461538y + 0.205462}$$

Ques from the following data obtain the two regression lines

X	y	x ²	y ²	xy
6	9	36	81	54
2	11	4	121	22
10	5	100	25	50
4	8	16	64	32
8	7	64	49	56
<u>30</u>	<u>40</u>	<u>220</u>	<u>340</u>	<u>214</u>

$$b_{xy} = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum y^2 - (\sum y)^2} = \frac{40(214) - (30)(40)}{40(340) - 115600} = -0.07215$$

$$b_{xy} = \frac{40(214) - 30 \times 40}{40(340) - 30 \times 40} =$$

$$\sum x = 30$$

$$\sum y = 40$$

$$\sum x^2 = 220$$

$$\sum y^2 = 340$$

$$\sum xy = 214$$

$$(\sum x)^2 = 900$$

$$(\sum y)^2 = 1600$$

$$n = 5$$

$$b_{ny} = \frac{5(214) - 30 \times 40}{5 \times 340 - 1000} = \frac{-0.1625}{-1.3}$$

$$b_{yx} = \frac{5(214) - 30 \times 40}{5 \times 220 - 1000} = -0.65$$

$$\bar{x} = 30/5 = 6$$

$$\bar{y} = 40/5 = 8$$

$$r = \sqrt{b_{ny} \times b_{yx}}$$

$$r = 0.919239$$

→ Y on X

$$(y - \bar{y}) = b_{yx}(x - \bar{x})$$

$$y - 8 = (-0.65)(x - 6)$$

$$y = -0.65x + 11.9$$

$$y = 11.9 - 0.65x$$

→ X on Y

$$(x - \bar{x}) = b_{xy}(y - \bar{y})$$

$$(x - 6) = -1.3(y - 8)$$

$$x = -1.3y + 16.4$$

$$x = 16.4 - 1.3y$$

- Multiple Linear Regression: x_1, x_2, x_3

$$x_1 = a + bx_2 + cx_3 \leftarrow \text{Regression plane}$$

[x_1 on x_2 & x_3]
The normal Equations are:

$$\sum x_1 = na + b \sum x_2 + c \sum x_3$$

$$\sum x_1 \cdot x_2 = a \sum x_2 + b \sum x_2^2 + c \sum x_2 \cdot x_3$$

$$\sum x_1 \cdot x_3 = a \sum x_3 + b \sum x_2 \cdot x_3 + c \sum x_3^2$$

- x_2 on x_1 & x_3 .

$$x_2 = a + bx_1 + cx_3$$

Normal Eq:- $\sum x_2 = na + b \sum x_1 + c \sum x_3$

$$\sum x_2 \cdot x_1 = a \sum x_1 + b \sum x_1^2 + c \sum x_1 \cdot x_3$$

$$\sum x_2 \cdot x_3 = a \sum x_3 + b \sum x_1 \cdot x_3 + c \sum x_3^2$$

- x_3 on x_1 & x_2

~~$$x_3 = a + bx_1 + cx_2$$~~

Ques Find the multiple linear regression equation of x_1 on x_2 & x_3 from the data related to three variables given below.

x_1	x_2	x_3	$x_1 \cdot x_2$	$x_1 \cdot x_3$	$x_2 \cdot x_3$	x_2^2	x_3^2
4	15	30	60	120	450	225	900
6	12	24	72	144	288	144	576
7	8	20	56	140	160	64	400
9	6	14	54	126	84	36	196
13	4	10	52	130	40	16	100
15	3	4	45	60	12	9	16
54	48	102	339	720	1034	494	2188

$$54 = 6a + 48b + 102c$$

$$339 = 48a + 494b + 1034c$$

$$720 = 102a + 1034b + 2188c$$

using Gauss elimination

$$\left[\begin{array}{ccc|c} 6 & 48 & 102 & 54 \\ 48 & 494 & 1034 & 339 \\ 102 & 1034 & 2188 & 720 \end{array} \right] \quad \begin{array}{l} \frac{a_2}{a_1} = 8 \\ \frac{a_3}{a_1} = 17 \end{array}$$

$$R_2 \rightarrow R_2 - 8R_1, \quad R_3 \rightarrow R_3 - 17R_1$$

$$\begin{bmatrix} 6 & 48 & 102 & 54 \\ 0 & 110 & 218 & -93 \\ 0 & 218 & 454 & -198 \end{bmatrix}$$

$$\frac{b_3}{b_2} = 1.981818$$

$$R_3 \rightarrow R_3 - 1.981818 R_2 \quad \frac{218}{110} R_2$$

$$\begin{bmatrix} 6a_1 & 48b_1 & 102c_1 & 54d_1 \\ 0 & 110 & 218 & -93 \\ 0 & 0 & 21.963636 & -13.690909 \end{bmatrix}$$

$$x = \frac{d_1 - c_1(x) - b_1(y)}{a_1}$$

$$y = \frac{d_2 - c_2(x)}{b_2}$$

$$z = \frac{d_3}{c_3}$$

$$a = 16.477649$$

$$b = 0.389902$$

$$c = -0.623345$$

$$x_1 = 16.477649 + 0.389902x_2 - 0.623345x_3$$

Estimate the value of x_1 when $x_2 = 4$ & $x_3 = 10$

$$x_1 = 11.803807$$

acc/ to ques expected
the value of $x_1 = 13$
when $x_2 = 4, x_3 = 10$

⇒ Non-Linear Regression:

$$y = ax + bx^2 + cx^3$$

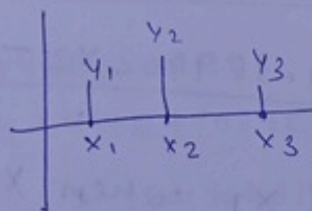
$$x = a + by + cy^2 \text{ (vice versa is possible)}$$

⇒ Frequency Distribution:

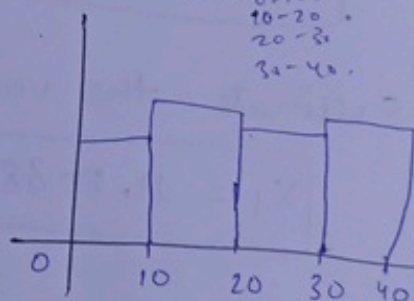
- (i) Univariate (single variable)
- (ii) Bivariate
- (iii) Discrete
- (iv) Grouped (Range)

⇒ Graphical Representation

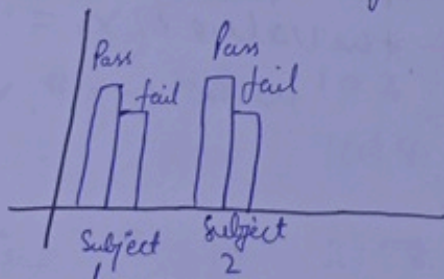
(i) Line Graph



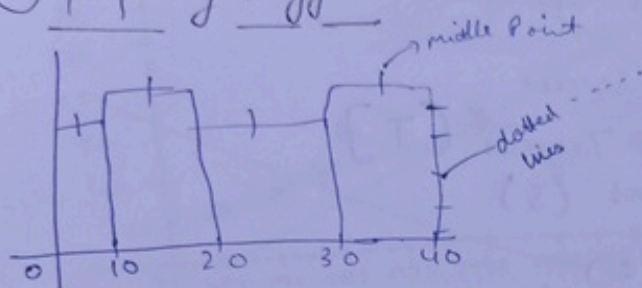
(ii) Histogram (Continuous Data)



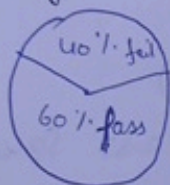
(iii) Bar Graph (Categorical Data)



④ Frequency Polygon:



⑤ Pie diagram.



eg:
 BCA 1st $\rightarrow$ 350
 BCA 2nd $\rightarrow$ 250
 BCA 3rd $\rightarrow$ 130

